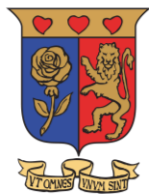


Sensors & Actuators

ICS 4111: Topic 5

Allan Vikiru



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Learning Objectives

- Recognise and compare the range of sensors and actuators used in IoT systems.
- Understand principles of interfacing hardware components for IoT applications.

Transducers



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Transducers are devices that convert one form of energy to another e.g. a microphone converts sound to electric signals

Type	Energy Conversion	Example
Electrochemical	Chemical to electrical energy and v/v	battery
Electroacoustic	Sound to electrical energy and v/v	microphone
Electromagnetic	Magnetic to electrical energy and v/v	antennas
Electrostatic	Static to electrical energy and v/v	Van de Graaff generator
Electromechanical	Motion/mechanical to electrical	piezoelectric
Photoelectric	Light to electrical	photodiode
Thermoelectric	Heat to electrical	thermocouple

Sensors

- Sensors are devices that respond to a signal. This signal is produced by some source of energy e.g. heat, light, motion.
- A sensor will convert this input signal to an analog or digital representation of the same signal.
- Classification of sensors:
 - Passive vs Active:
 - **Passive:** do not require an external power source like a mercury thermometer
 - **Active:** require a power source e.g. DHT22 temperature and humidity sensor

Sensors

- Classification of sensors:
 - Analog vs Digital:
 - **Analog:** produce continuous signals synonymous to the sensed parameter like a sound sensor. Will need conversion before interfacing to the digital controller.
 - **Digital:** generates digital signals (0 for absence and 1 for presence of measured parameter). Example: infrared sensor, does not need conversion when interfacing with controller.

Sensors



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Type	Function	Example
Thermal	Detects level of heat	LM35, DHT22
Mechanical	Detects motion	ADXL345
Electrical	Detects electrical energy	Voltage sensor
Chemical	Detects type/quantity of chemicals	MQ-2
Optical	Detects light	BPW34
Radiation	Detects radiation	GDK101
Acoustic	Detects sound waves	KY-038

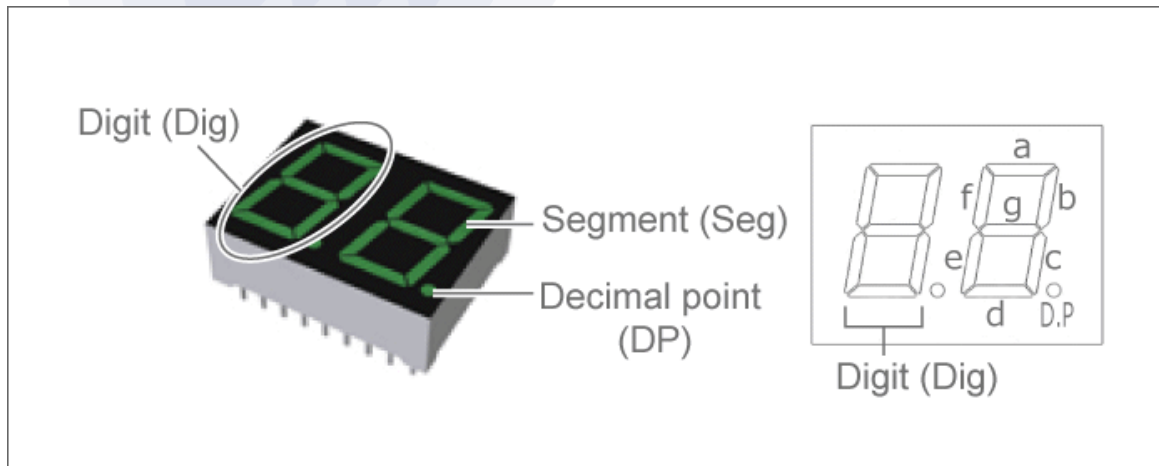
Actuators

Actuators are devices that uses energy to provide motion.

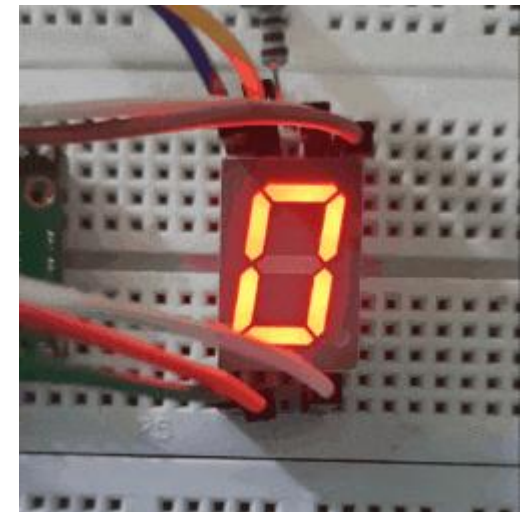
Type	Function	Example
Thermal	Converts thermal energy to motion	bimetallic strip
Electric	Converts electrical energy to motion	motor
Mechanical	Converts mechanical energy to motion	screw jack

Actuators – I/O Components

1. **Seven-segment LED:** a digital display module to display numerical or text information. LEDs are arranged in the shapes of numbers to provide a visible display

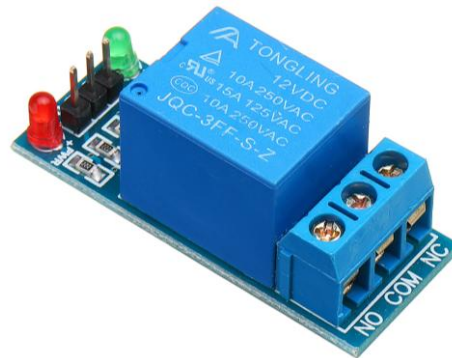


(Source: [ROHM Semiconductor, 2025](#))



Actuators – I/O Components

2. **Relay:** devices used to control circuits that operate with low-power signals. It has two metal contacts and a switch such that the switch connects these contacts and allows for current flow. This feature makes them suitable for load balancing in large projects.



(Source: [Leung, 2025](#))

Actuators – I/O Components



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3. **Piezo Buzzer:** a device used to produce a tone, alarm or sound.



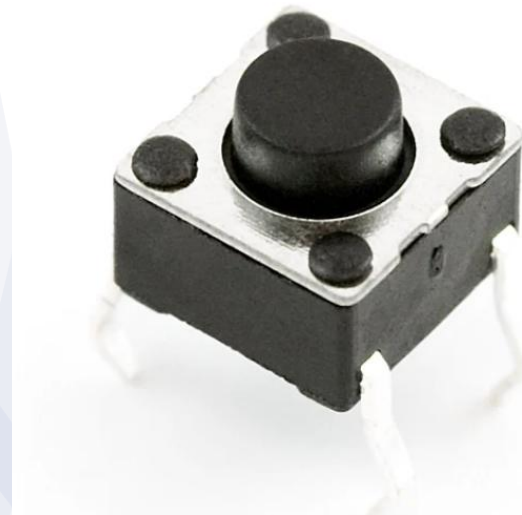
(Source: [Kendall, 2023](#))

Actuators – I/O Components



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4. **Push Button Switch** : a device used to close or open a circuit when pressed.



(Source: [Sparkfun Electronics, 2025](#))

That's Part 1!

Any Questions?



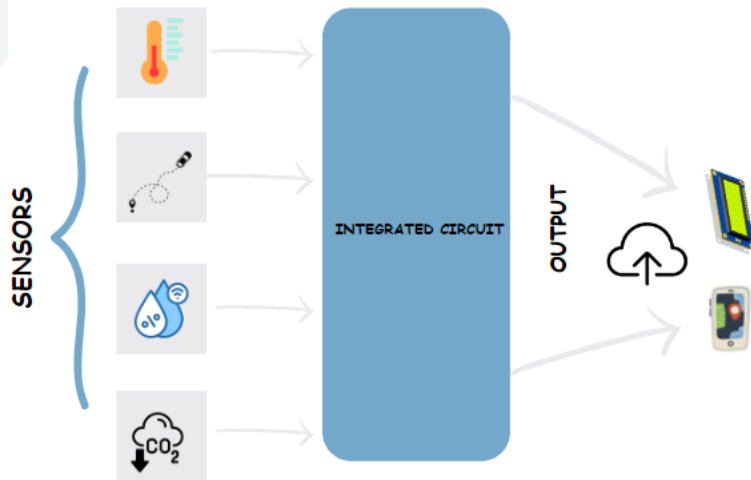
Scan this code as you take a quick break



Part 2: Principles of Interfacing!



Interfacing



(Source: [Watson, 2024](#))

- Interfacing involves connecting sensors, actuators and a controlling unit.
- In developing embedded systems, it is essential to validate:
 - the measurements that will be captured, and how they will be presented,
 - the type of signal of input & output (digital/analog),
 - power supply throughout the circuit,
 - how each component functions,
 - compatibility of components,
 - application area of the system.



General Requirements for Interfacing

- **Input signal conditioning:** involve setting the offset and scaling the signal properly from input devices
- **Output signal conditioning:** involve setting the level/power of the signal for the connected devices
- **Noise cancellation:** involves taking care of surrounding interference signals
- **Error compensation:** taking care of errors introduces by A/D & D/A

Performance Characteristics of Sensors & Actuators



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- **Accuracy** (*correctness of the output*) – closeness between measured value and true value
- **Precision** (*reliability or reproducibility of output*) – degree of consistency among measurements of same quantity
- **Error margin** – percentage of confidence level in correctness of output

Performance Characteristics of Sensors & Actuators



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- **Nonlinearity:** relationship between output signal and input signal
- **Saturation point:** how far before input has no effect on output
- **Sensitivity analysis:** determine how much output varies with a change in input parameters.
- **Calibration and resolution:** adjusting the performance of a component to determine if it can capture the smallest possible change.

Performance Characteristics of Sensors & Actuators



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- **Application environments** (i.e. heat, cold, rugged, humidity, dryness)
- In determining/analysing the performance of components for embedded systems, engineers refer to their datasheets. (see [this](#) example of a DHT22 sensor)



Tools for Interfacing

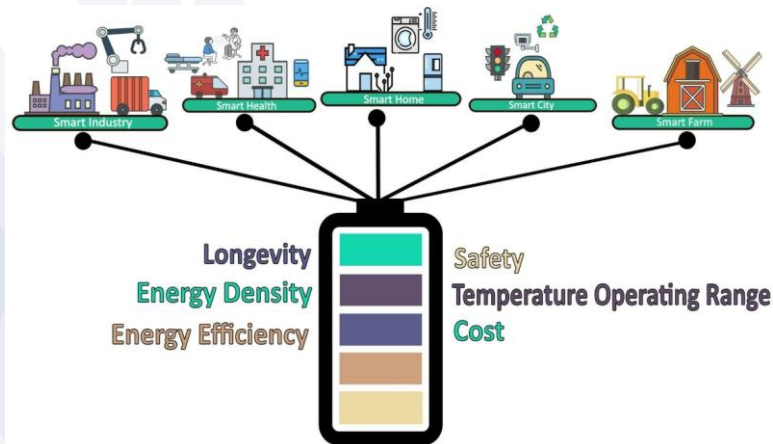
- **Amplifiers** – increases amplitude/power of a signal. Usually DC energy.
- **Repeaters** – device that receives a signal and retransmits it.
- **Converters** – changes signal from one form to another to enhance integration
- **Noise cancellation, signal isolation, signal compensation** to increase strength of signal.

Powering Embedded Devices



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- Embedded devices are constrained in nature: components typically operate between 3.3 and 5V, with currents between 30 mA to 1A.
- Mains power supply can be as high as 240V, which can damage the device. It is therefore it is crucial to verify the power supply requirements.



(Source: [Hasan et al, 2023](#))

Powering Embedded Devices



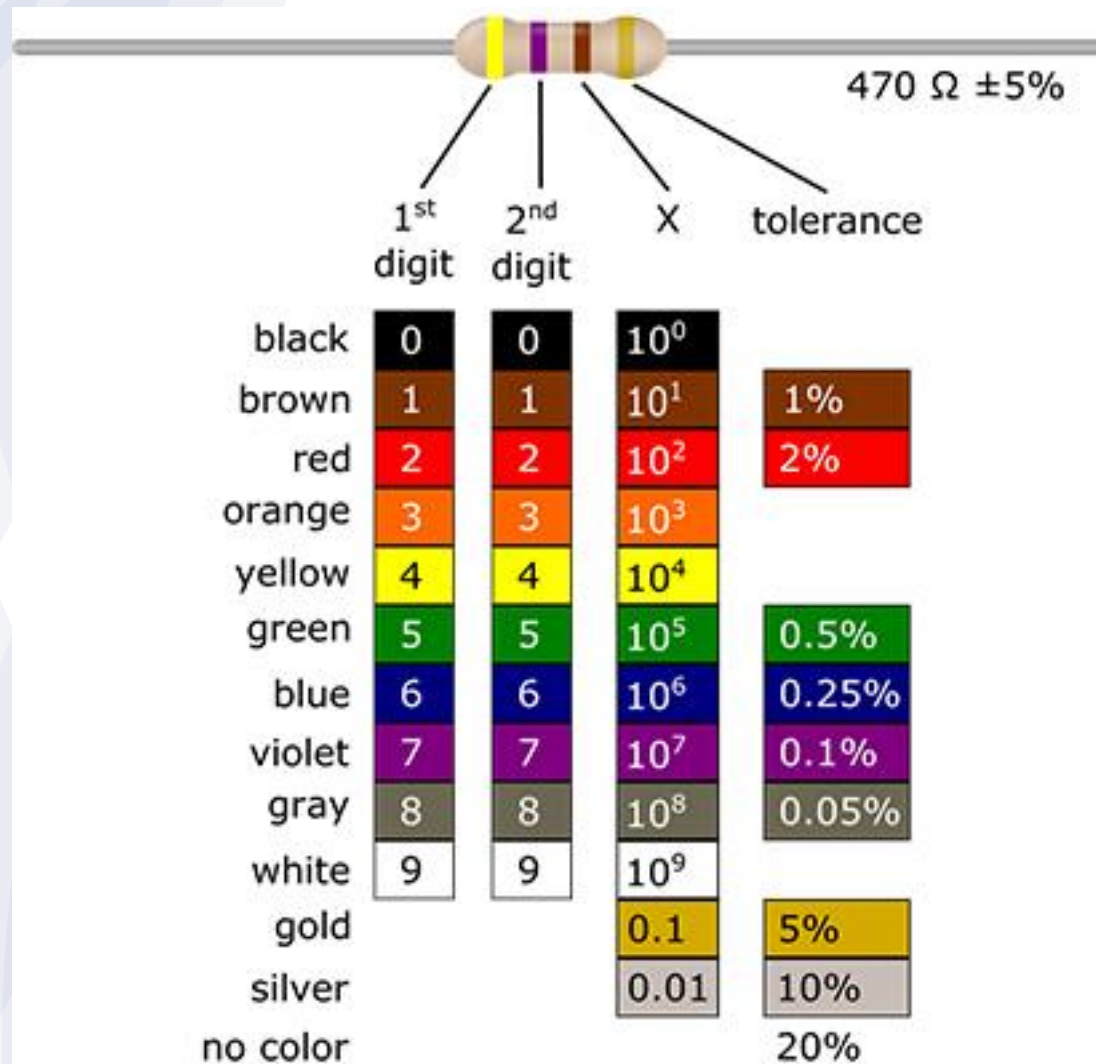
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- In developing IoT architectures, the **power budget** is calculated to determine if the power requirements are satisfactory.
- Common types of power supplies for embedded devices: *lithium-ion batteries* (rechargeable), *alkaline batteries* (non-rechargeable) and *lead acid batteries* (for structurally large devices)
- By default, MCUs have a ground ports to return excess power back to the power supply.
- Resistors are also implemented in circuits to control the flow of current.

Reading Resistors



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Source: ([John Wiley & Sons Inc, 2016](#))



That's all for now! Any Questions?

Next – Topic 6: System Prototyping